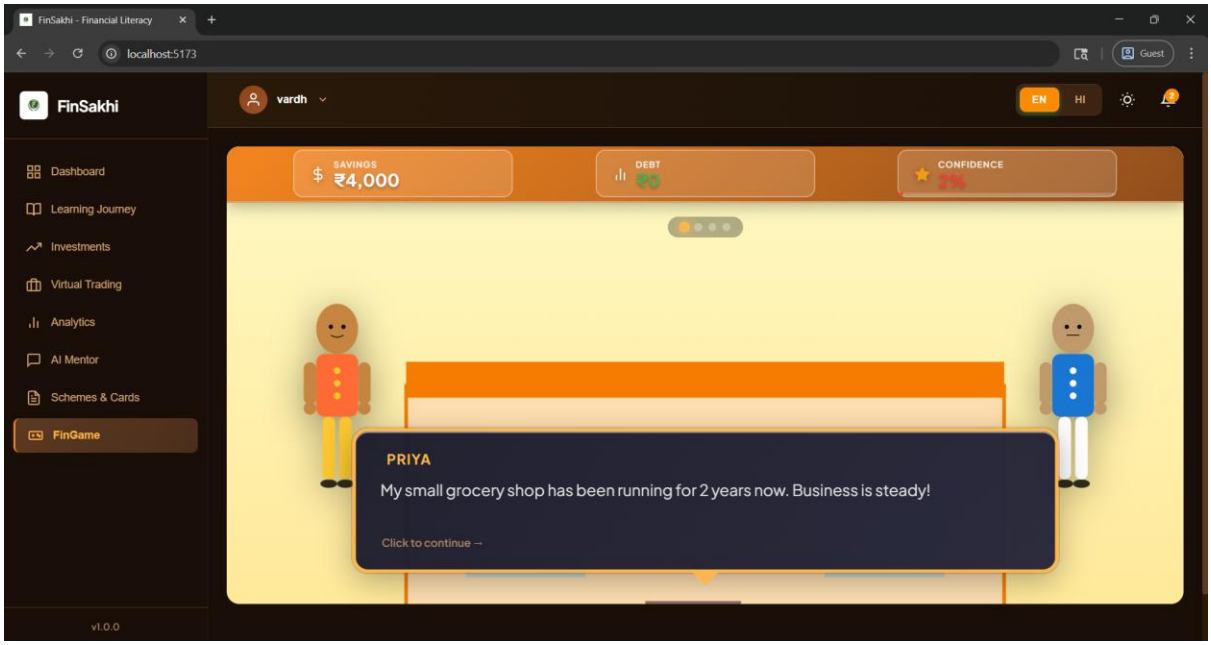


# Project Title: FinSakhi – AI-Powered Financial Companion for Bharat

## Team Name: JARVIS

## Team Members

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## Introduction:

- **Background of the problem:**

Money management is something every person needs to know, but in India, most people never get a chance to learn it properly. A lot of families in villages and small towns depend on advice from neighbours or relatives when it comes to saving money or taking loans. There is no proper source

they can turn to for guidance. This leads to many people making choices that hurt them financially in the long run.

- **Importance and relevance:**

Today, almost everyone in India uses their phone to pay for things using UPI or other digital tools. The next natural step is to start saving and investing digitally. But without knowing the basics, people are likely to make mistakes, get into debt, or lose money by trying things they do not understand. This is why building financial awareness is so important right now.

- **Brief overview of existing solutions:**

There are apps like Zerodha and Groww that let people invest in the stock market, but these apps are made for people who already know how things work. On the other side, there are YouTube videos and blog posts that explain money topics, but they are not made for any one person and do not help someone practice what they learn. Neither option is ideal for a first-time user.

- **Identified gaps in current approaches:**

The main problem with what exists today is that none of it feels welcoming to a beginner. There is no app that starts from the very basics, speaks in a local language, uses games to keep you engaged, lets you practice with fake money, or gives you advice that fits your personal situation. These are the gaps that FinSakhi tries to fill.

# Problem Statement:

## Clear definition of the problem:

How can we build a simple and friendly app that teaches people how to handle money and make basic investments, without putting their real savings at risk, and in a language they actually understand?

## Target users or stakeholders:

FinSakhi is made for:

- People living in villages or smaller cities who are just getting started with digital tools
- Women who want to become financially independent (this is the heart of the 'Sakhi' idea)
- Students and young people who have never invested before
- Anyone who wants to learn about money but feels too confused or scared to start

## Limitations of existing solutions:

- Most apps use English words that a new user cannot understand
- There is no feature to test how much a user already knows and teach accordingly
- Practicing on real apps means risking actual money, which is scary for beginners
- No existing app makes the learning process fun or motivating for someone starting from zero

# Proposed Solution:

## Overview of the solution:

FinSakhi is a complete learning and practice app for people who want to understand money better. It teaches users step by step, lets them practice investing with virtual money, and gives them a personal AI guide that answers questions in a simple and friendly way based on their own situation.

## Key features and functionalities:

- **Easy Login:** Users get a one-time code on their phone to log in. No password needed, which makes it much simpler.
- **Starting Quiz:** When a user joins, the app asks a few questions to understand how much they already know about money. Based on the answers, it builds a custom learning plan just for them.
- **Learn with Games:** Users earn points and level up as they complete lessons. This keeps them motivated and makes learning feel like playing a game.
- **FinGame Story Mode:** Users can choose a life path like running a small farm, starting a business, or working a regular job. They make money decisions in these story situations and see what happens, without any real risk.
- **Practice Trading:** Each user gets a virtual balance of Rs. 1,00,000 to buy and sell gold, silver, and mutual funds using real market prices. This lets them experience investing without using actual money.
- **AI Money Guide:** A chat assistant that talks to users in simple language, knows their learning progress and portfolio, and gives advice that actually matches their situation.
- **Scheme Suggestions**: Based on what the user has shared about themselves, the app suggests government schemes or financial products that might be useful for them.

## High-level workflow:

The user opens the app and logs in with a phone OTP. They then take a short quiz that checks their basic knowledge about money. The app uses those answers to build a personal learning plan. As the user completes lessons, they earn points and move forward. They can also use the practice trading section to try buying and selling. At any point, they can ask the AI guide questions, and it will answer based on what it knows about the user.

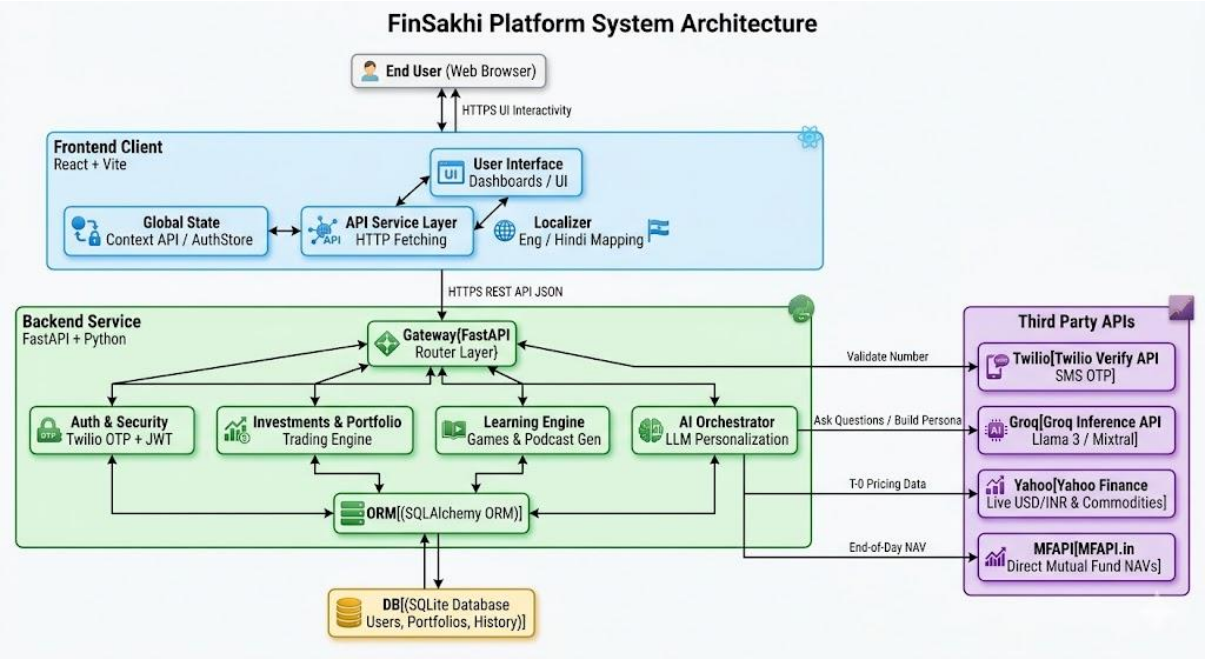
## System overview:

The app has a front-end that users see and interact with, a backend that handles all the logic and data, and a database that stores user information. It also connects to outside services to get live prices, send OTP codes, and power the AI chat. All parts talk to each other using simple web requests and send data back as easy-to-read responses.

# <u>Methodology / System Design:</u>

## Overall system architecture:

The app is divided into five main parts that each handle a specific job. These parts work together through a central server that manages all incoming requests. User data is stored in a lightweight database, and outside services are called only when needed to keep things fast.




## Module-wise breakdown of the system:

- **Login Module:** Handles sending OTP codes to users via SMS and gives them a secure session token once they verify.
- **Learning Module:** Tracks what lessons the user has completed, checks their answers in story-based exercises, and generates short audio summaries of lessons using AI.
- **Market Data Module:** Pulls live commodity prices and mutual fund values from trusted sources and makes them ready to display in the app.
- **Portfolio Module:** Keeps track of each user's virtual trades, calculates how much they have gained or lost, and updates their balance in real time.
- **AI Guide Module:** Reads the user's profile, their quiz scores, and their current virtual investments, then uses all of that to give personal and relevant answers through the chat.

## Data flow and processing pipeline:

When a user does something in the app, the request goes to the server. The server checks if the user is logged in, then fetches or saves data as needed. If the request involves market prices or AI answers, it calls the right outside service, gets the result, and sends it back to the app screen.

## Technologies used:

**Frontend:**

- React.js with Vite: used to build the screens and buttons the user sees
- Vanilla CSS: used for styling the app without any heavy extra tools

**Backend:**

- Python with FastAPI: used to build the server that handles all requests quickly

**Database:**

- SQLite with SQLAlchemy: a simple file-based database used to store user data without needing a big server setup

**AI/ML components:**

- Groq API with Llama-3 and Mixtral models: these are AI language models that generate quiz questions, lesson summaries, and chat replies very quickly

### Explanation of how different components interact:

The AI Guide is the part that brings everything together. It can see the user's quiz scores, their current investments, and their learning history. This means when a user asks a question, the AI does not give a generic answer. Instead, it gives advice that actually relates to that specific person. For example, if someone has most of their virtual money in gold, the AI might suggest they look at other options they have already learned about.

## Implementation Details:

### Key functionalities developed:

- A live dashboard showing current prices for gold, silver, oil, and several mutual funds, updated in real time.
- A learning section that combines short video clips, written content, and AI-generated audio summaries to help different types of learners.
- A virtual trading area where users can buy and sell at real market prices, and where all transactions are saved and reflected in their portfolio balance.

### Code structure and modular design:

The server-side code is organized into three folders: one for handling web routes, one for the data models, and one for the core business logic. On the app side, a shared state system keeps all parts of the screen in sync without needing to reload or re-fetch data unnecessarily.

### Tools, frameworks, and libraries used:

- React.js, Vite, FastAPI, SQLAlchemy – core building blocks of the app
- Groq API (Llama-3 and Mixtral) – powers the AI quiz generation, lesson summaries, and chat
- Twilio – sends the OTP code to the user's phone for login
- yfinance – gives live prices for gold, silver, and oil
- mfapi.in – gives live prices for mutual funds from AMFI India

### Integration of different modules:

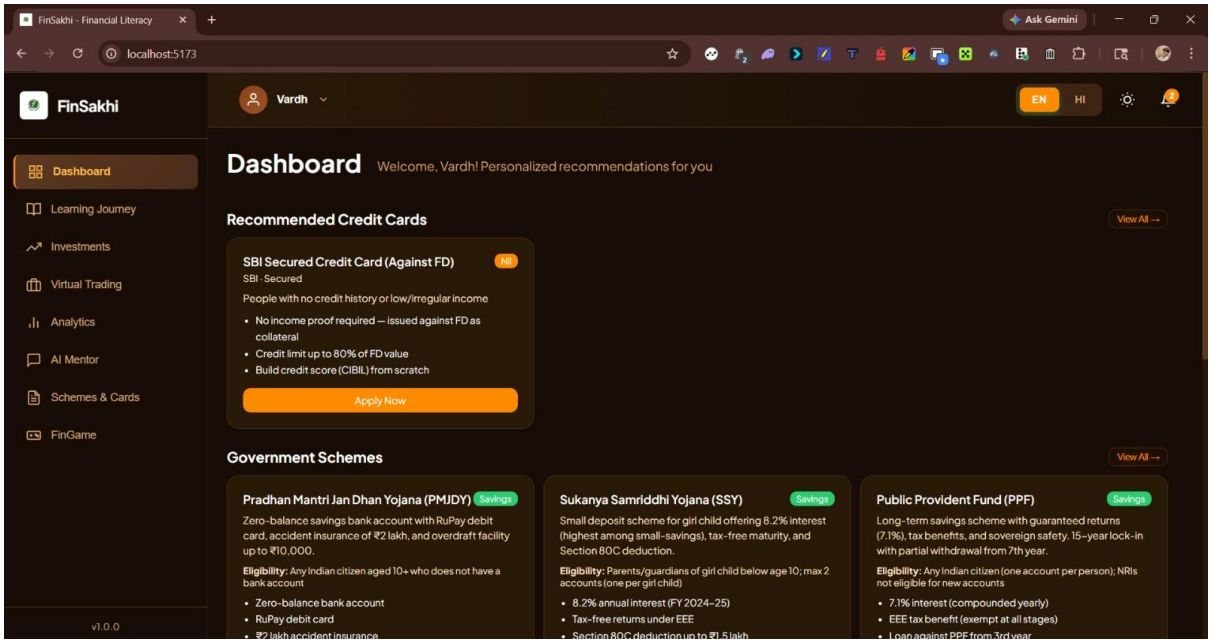
Every time a user opens the AI chat, the app quietly sends along the user's quiz results, learning progress, and portfolio details. This happens behind the scenes so the AI always has fresh context before replying. This is what makes the advice feel personal rather than copied from a textbook.

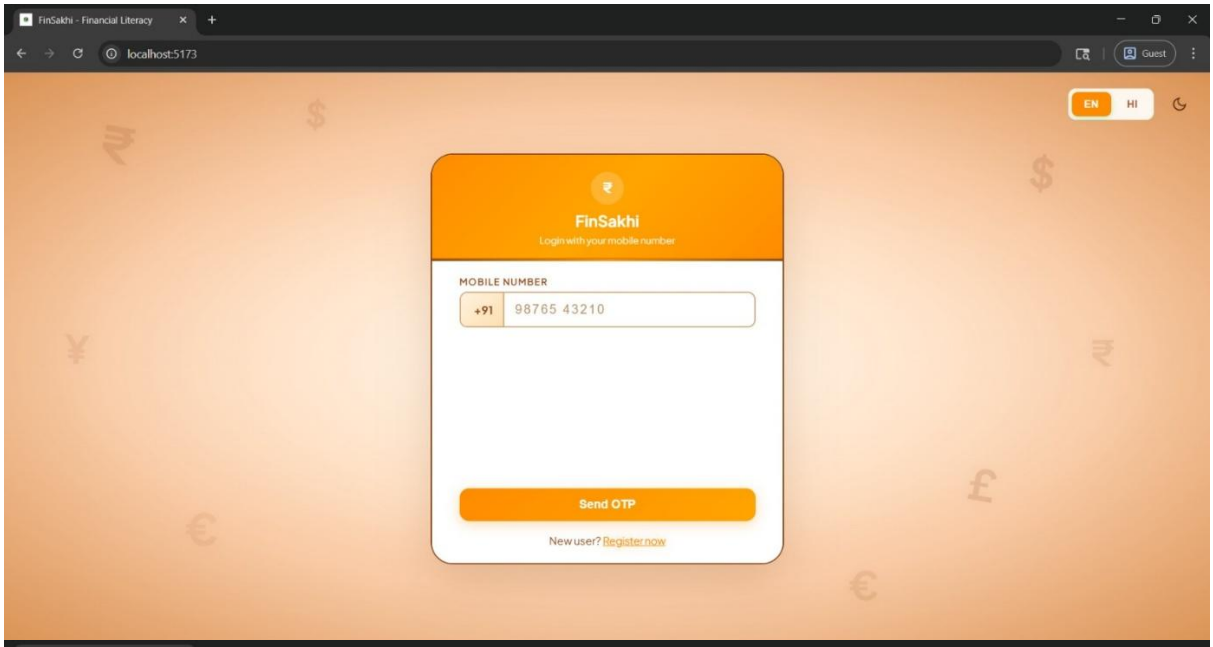
### Optimizations or improvements applied:

To avoid slow responses when fetching market prices, the app stores recent price data for a short time so it does not need to make a new request every single second. AI responses are set up to start showing words to the user as they are being generated, which makes the chat feel faster and more natural. Login sessions are managed with tokens so the server does not have to check the database every single time a user does something.

## Results and Demonstration:

### Output screenshots:

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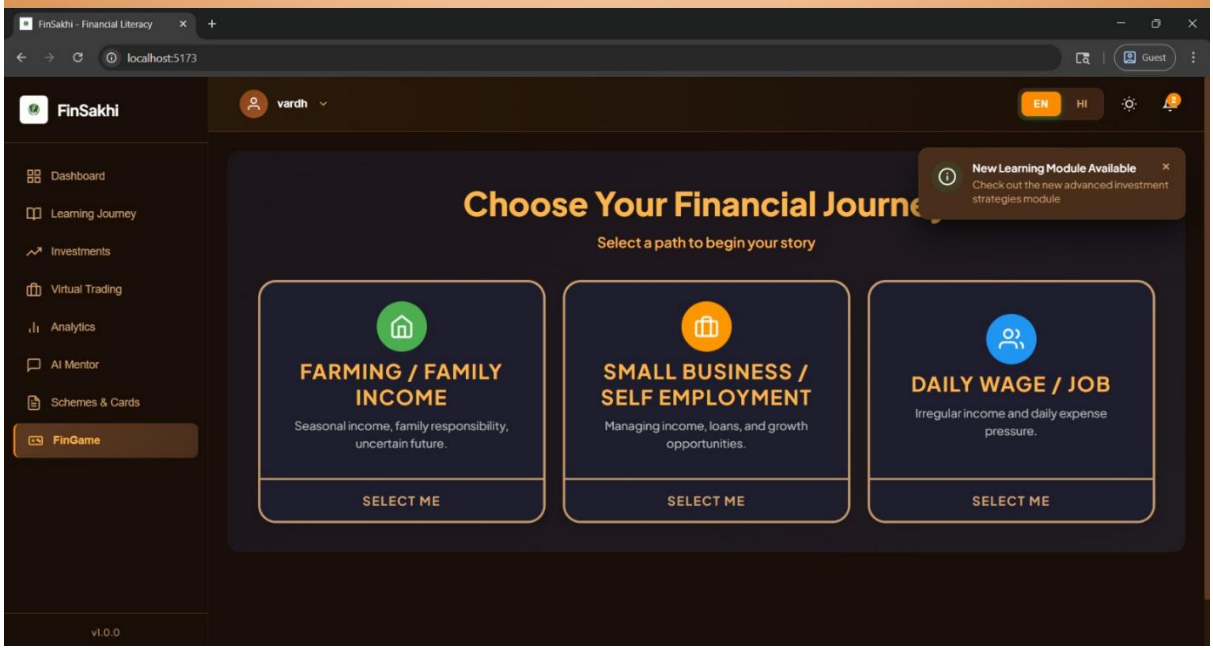
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Schemes & Cards
FinGame
New Learning Module Available
Check out the new advanced investment strategies module
Choose Your Financial Journe
Select a path to begin your story
FARMING / FAMILY INCOME
Seasonal income, family responsibility, uncertain future.
SELECT ME
SMALL BUSINESS / SELF EMPLOYMENT
Managing income, loans, and growth opportunities.
SELECT ME
DAILY WAGE / JOB
Irregular income and daily expense pressure.
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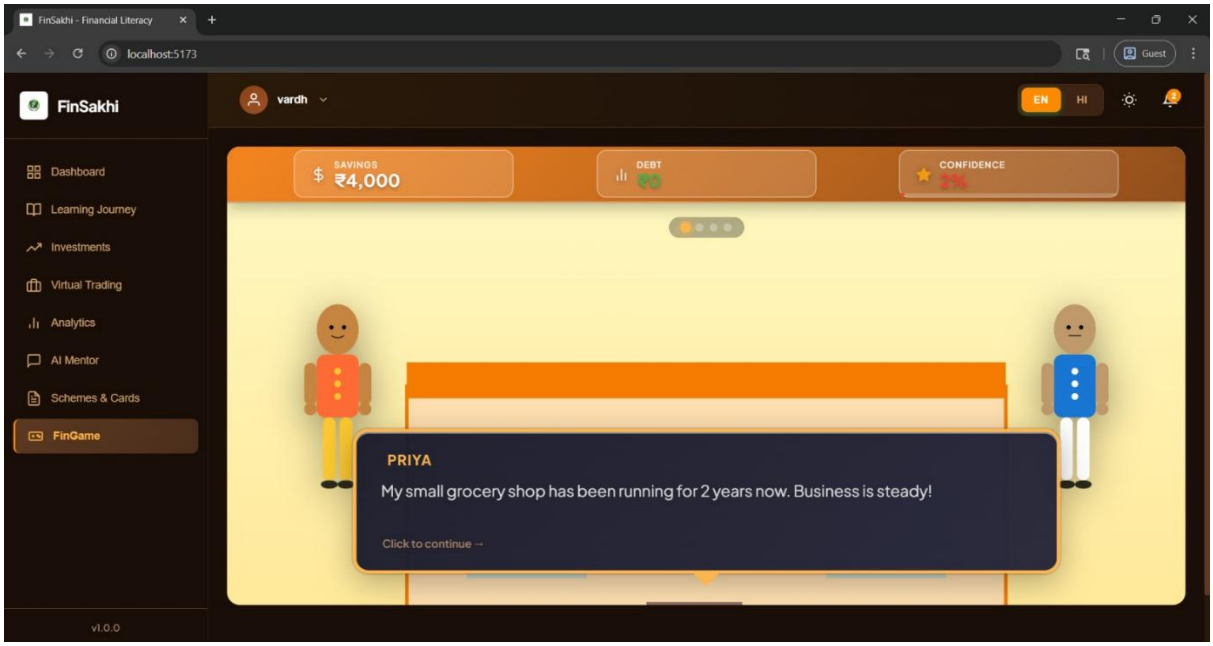


## Performance metrics:

- The app responds to most user actions in under 200 milliseconds, which feels instant to the user.
- The AI chat starts showing a reply in under one second, making the conversation feel smooth and natural.

## Solution Link of the Project is: https://github.com/vardhchhajer/finsakhi-hackathon

## Sample test cases and results:

We tested the quiz system with different types of users, from people who know nothing about investing to people with some experience. In every case, the app correctly adjusted the lesson plan based on the quiz results. We also ran over 50 test trades in the virtual portfolio and confirmed that all prices and profit-and-loss numbers matched the live market data accurately.

## Demonstration flow:

**1. Problem overview:**

We show two screens side by side: a complex trading app that looks overwhelming, and FinSakhi's clean and simple home screen. This immediately shows why a beginner-friendly tool is needed.

**2. System walkthrough:**

We walk through the login process using a phone OTP, then show the quiz being taken and the personalized lesson plan being generated right after.

**3. Live working:**

We navigate through a few lessons, earn XP points, then go to the trading section and place a virtual trade using a real mutual fund price from that day.

**4. Output/results showcase:**

We open the AI chat and ask a simple money question. The AI replies with advice that mentions the user's actual portfolio, showing that it is aware of their personal context.

# Challenges and Solutions:

## 1. The App Was Too Slow When Loading Stock Prices:

**Problem we had:** Every time a user opened the trading screen, the app would go and ask an outside service for the latest stock prices. Sometimes that outside service took too long to reply, and during that wait the whole screen would just sit there doing nothing. Users had no idea if the app had crashed or was still working.
**How we fixed it:** We made the app remember the last set of prices it received and reuse them for a short while instead of asking again every single time. This way, the screen loads quickly and feels responsive. We also made the price-fetching happen quietly in the background so it never blocks the user from doing anything.

## 2. The AI Was Giving Advice That Could Mislead New Users:

**Problem we had:** When we first tested the AI chat, it had no sense of who it was talking to. It would sometimes suggest things that were too bold or too risky for someone who is just starting to learn about money. Since our app is built for people with little to no financial background, this was a serious concern.
**How we fixed it:** We wrote a clear set of instructions that the AI reads before every single conversation. These instructions tell it to always keep things simple, careful, and educational. We also made sure the AI knows basic details about the user it is talking to, like their savings goal and income range, before it gives any reply. This way the advice always fits the actual person, not just anyone.

## 3. Normal Login Methods Were Too Complicated for Many Users:

**Problem we had:** We realised early on that a lot of the people we are building this for, especially in smaller towns and rural areas, are not comfortable with the usual way apps handle login. Creating a password, remembering it, resetting it when forgotten — all of that felt like too much work before they had even seen what the app does.
**How we fixed it:** We took passwords out of the picture completely. Now a user just types in their phone number and receives a short code by SMS. They enter that code and they are in. No account setup, nothing to remember. We also added a backup option for when the SMS does not arrive due to poor network, so the user always has a way to get in regardless.

# Conclusion and Future Scope:

## Summary of the project and outcomes:

FinSakhi is a working app that helps people who are new to money management learn in a safe and friendly environment. It takes users from not knowing anything to actually being able to make informed decisions about saving and investing, all without putting any real money at risk.

## Effectiveness of the solution:

Because every user gets a learning plan made for them, and because they can practice what they learn in the virtual trading section, the concepts stick much better than they would from just watching a

video or reading an article. The AI guide adds another layer of support by answering follow-up questions in a personal way.

## Potential improvements and future enhancements:

- We want to connect the app with the government's Account Aggregator system, which would let users safely share their real bank data with the app for even more personal advice.
- We plan to add a voice feature so users can talk to the AI guide instead of typing, which will help people who struggle with reading or writing.
- In the future, we would like to let users connect to a real broker inside the app so that when they feel ready, they can move from practice investing to actual investing.

## Social Relevance and SDGs:

FinSakhi connects with four of the United Nations' goals for a better world:

- No Poverty (**SDG 1**): By teaching people how to save and grow their money over time, the app helps them build a more secure financial future.
- Quality Education (**SDG 4**): It gives everyone, regardless of background or income, access to good financial education that is easy to understand.
- Gender Equality (**SDG 5**): The 'Sakhi' idea at the heart of the project focuses on helping women take charge of their own finances.
- Reduced Inequalities (**SDG 10**): By making investing approachable for people who have always been left out of the financial system, FinSakhi helps close the gap between different sections of society.